

Experimental Evaluation of a Multi-Layer Abstraction Approach for RL on Non-Markovian Tasks

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Outline

- ▶ Motivation
- ▶ Framework and Implementation
- ▶ Experimental Evaluation
- ▶ Conclusions



Motivation and Problem Statement

1 Motivation

Sparse Rewards

Informative feedback
is received only after
reaching the final goal.

Non-Markovian Task

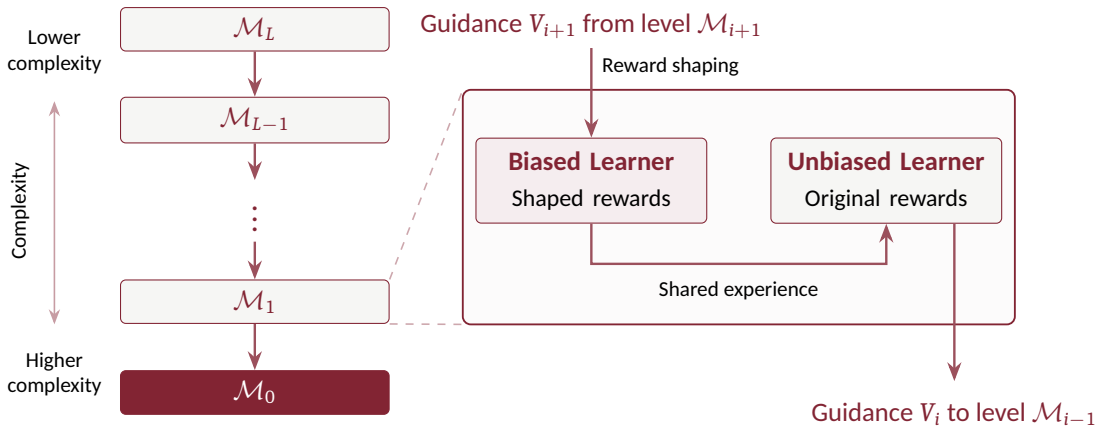
Task completion may depend
on the history of previously
observed events.

Prohibitive Learning



The Starting Point

2 Framework and Implementation

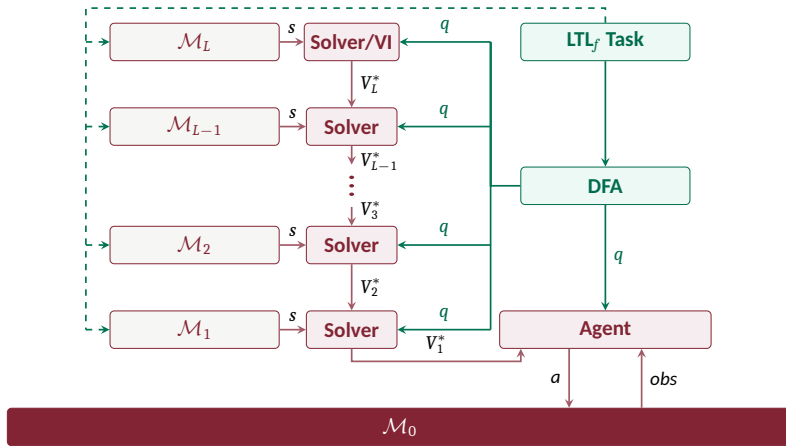


Framework by Cipollone et al. [4]



Extension Proposed in This Thesis

2 Framework and Implementation



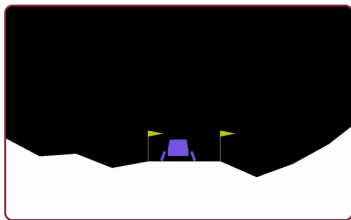


Experimental Setting

3 Experimental Evaluation

Ground Domain M_0

Gymnasium Lunar Lander



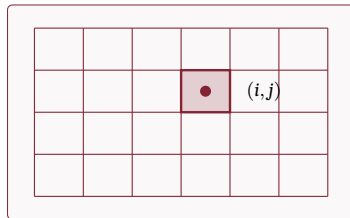
- $S = (x, y, v_x, v_y, \theta, \omega, c_l, c_r)$
- $\mathcal{A} = \{\text{no-op, eng-left, eng-right, eng-main}\}$
- Original dense reward removed

State mapping



Abstract Domain M_i

Grid Environment



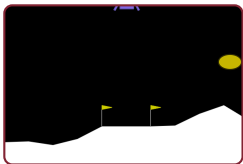
- $\mathcal{S}_i = \{(i,j) : 1 \leq i \leq H, 1 \leq j \leq W\}$
- $\mathcal{A} = \{\uparrow, \nearrow, \rightarrow, \searrow, \downarrow, \swarrow, \leftarrow, \nwarrow\}$
- Physical quantities discarded



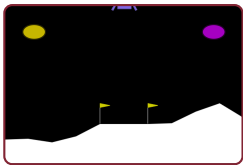
Biased-Unbiased Interaction

3 Experimental Evaluation

Episodic Tasks

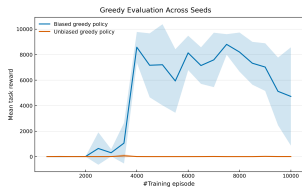
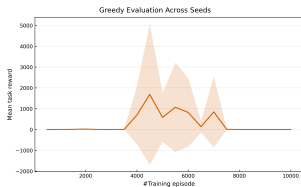
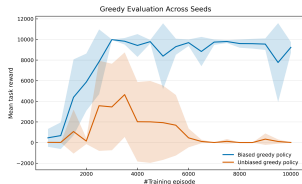
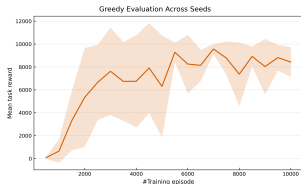


1-phase: $\diamond g_1$



2-phase: $\diamond(wp_1 \wedge \bigcirc \diamond g_1)$

Training results



Greedy eval.: standalone unbiased (left), coupled biased/unbiased (right); 1-phase (top) and 2-phase (bottom).



Temporal Misalignment

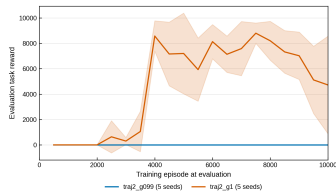
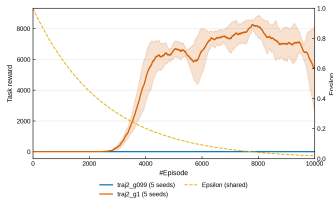
3 Experimental Evaluation

Shaping Formulation

$$\begin{cases} F(s, s') = \gamma_s \Phi(s') - \Phi(s) \\ \Phi(s) = V(s) \end{cases}$$

Potential Based Reward Shaping (PBRS) [4, 14].

Training Results



γ_s	Best success rate (%)	Same-state fraction (%)	Same-state reward per transition	Same-state reward per episode
0.99	0.00 $\pm$ 0.00	82.39 $\pm$ 0.96	-89.51 $\pm$ 0.05	-5000.60 $\pm$ 53.21
1.00	96.94 $\pm$ 1.91	91.48 $\pm$ 0.22	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00

Training reward (left) and greedy evaluation (right), with transition statistics for $\gamma_s = 0.99$ and $\gamma_s = 1$.



From Theory to Practice

3 Experimental Evaluation

Theory

Coupled architecture

Biased + Unbiased

PBRs Formulation

$$F(s, s') = \gamma_s V(s') - V(s)$$

Practice

Biased Learner

Only biased on the ground MDP

Shaping Heuristic

$$F(s, s') = V(s') - V(s)$$

Interaction
Problem

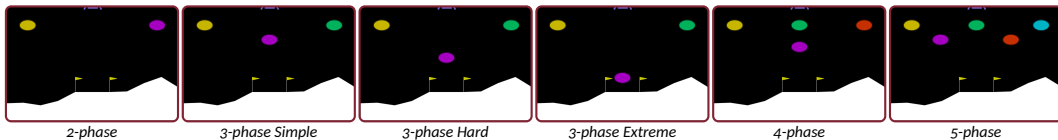
Temporal
Mismatch



Scaling to More Complex Non-Markovian Tasks

3 Experimental Evaluation

Required Tasks



Training Results

Task	Baseline	6×6	12×12	24×24	48×48	180×180	360×360
2-phase	16.1	91.9	97.4	96.8	98.3	95.1	97.0
3-phase Simple	5.3	95.0	94.1	98.2	98.3	98.8	96.5
3-phase Hard	0.1	95.0	93.1	97.0	88.0	98.6	97.1
3-phase Extreme	0.0	28.3	77.8	93.2	87.6	90.1	91.6
4-phase	0.0	24.8	53.5	83.6	74.9	64.2	85.1
5-phase	0.0	52.3	62.3	65.2	87.3	88.5	85.6

Best-policy success rate (%), mean over 10 training seeds.



Learning Reliability

3 Experimental Evaluation

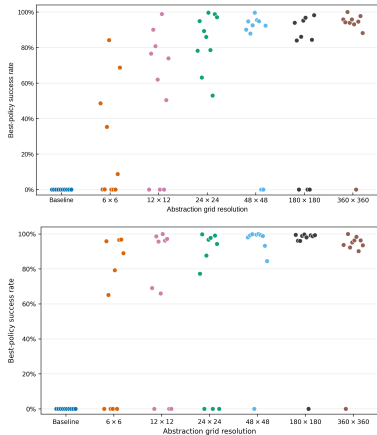
Reliability vs. Quality

Grid	4-phase $N \geq 80\%$	4-phase $SR \geq 80\%$	5-phase $N \geq 80\%$	5-phase $SR \geq 80\%$
6×6	1/10	84.2%	4/10	94.6%
12×12	3/10	89.9%	5/10	97.5%
24×24	6/10	94.3%	6/10	95.9%
48×48	8/10	93.4%	9/10	96.9%
180×180	7/10	91.2%	9/10	98.4%
360×360	9/10	94.8%	9/10	95.1%

$N \geq 80\%$: number of high-performing seeds.

Mean $SR \geq 80\%$: mean post-training best-policy success rate among high-performing seeds.

Seed-Level Distributions



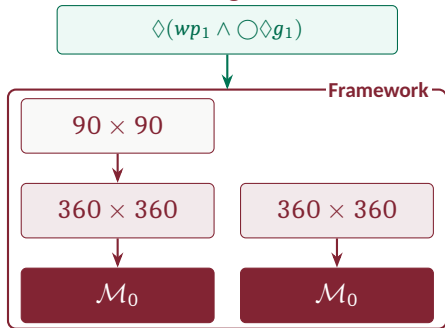
Top: 4-phase trajectory; bottom: 5-phase trajectory.
Each point represents one training seed (10 seeds per configuration).



Learning on Abstract Levels

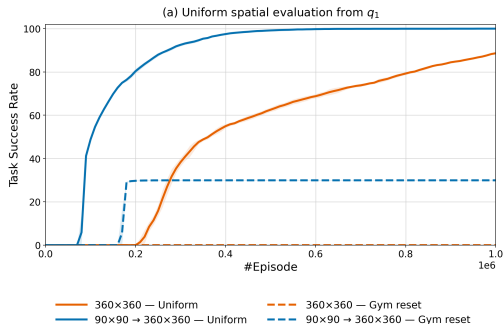
3 Experimental Evaluation

Learning Stacks



$90 \times 90 \rightarrow 360 \times 360 \rightarrow \mathcal{M}_0$ (left) VS $360 \times 360 \rightarrow \mathcal{M}_0$ (right)
on the 2-phase task.

Training Results



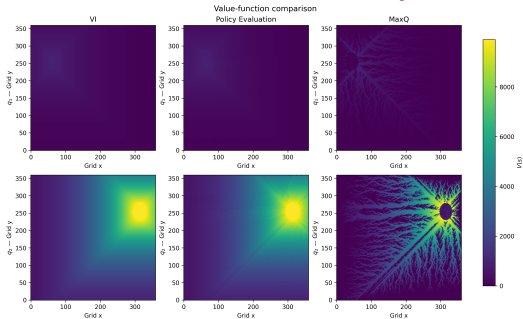
Task success rate during training under uniform and Gym-reset evaluation.



From a Learned Policy to an Effective Shaping Potential

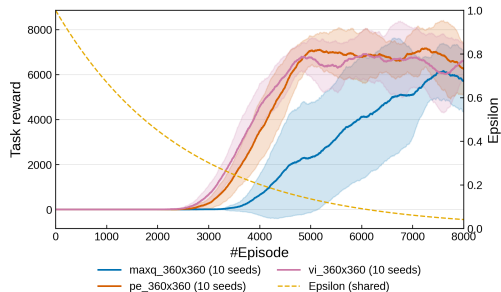
3 Experimental Evaluation

Value-Function Heatmaps



Value functions for VI, Policy Evaluation, and MaxQ (top: q_1 ; bottom: q_2 automaton states).

Training Results



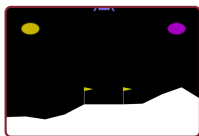
Task reward during training.



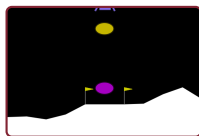
Learning on Cyclic Tasks

3 Experimental Evaluation

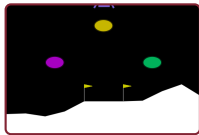
Cyclic Tasks



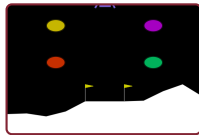
Horizontal



Vertical

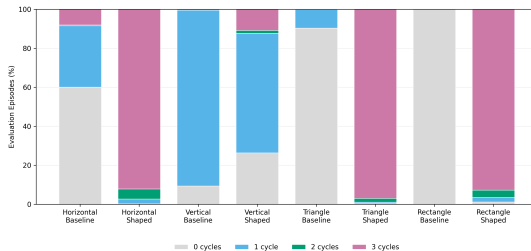


Triangle



Rectangle

Cycles Distribution



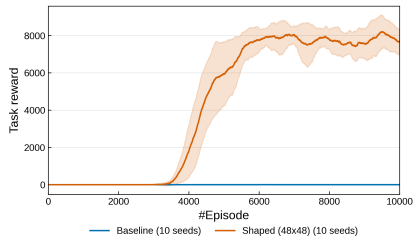
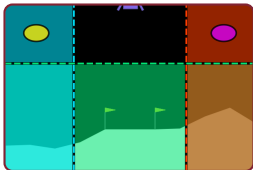
Evaluation episodes grouped by completed cycles (0-3), for baseline and shaped policies.



Additional Case Studies

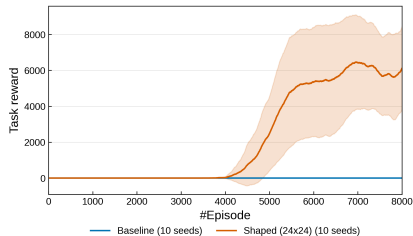
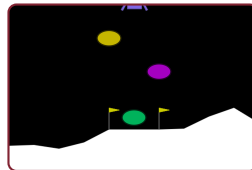
3 Experimental Evaluation

Task with Constraints



Task reward during training of 2-phase with constraints.

Landing Task



Task reward during training of landing task.



Findings, Limitations & Future Works

4 Conclusions

Findings

- Effective guidance extracted from abstractions
- Automatic guidance generation starting from logical formulas
- Multi-level architecture is promising

Limitations

- Coupled architecture interaction
- Temporal misalignment between levels
- Abstract policies could be not effective as guidance
- Long temporal tasks are difficult to be discovered

Future Works

- Study new exploration and experience-related mechanism
- Introduce different techniques across levels (e.g. planners)
- Richer logical formalism (e.g. STL [12])



Thank you for listening!
Any questions?



References

4 Conclusions

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